Predicting Customer Churn

Comparing Six Classification Model

26.5%

of customers in this dataset churned

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Approach

70 / 15 / 15 Stratified Split

Train, validation, and test sets each preserved the same 26.5% churn rate, so no split was accidentally imbalanced relative to the others.

SMOTE Balancing

Fake minority oversampling was applied to the training set only, never the validation or test, so evaluation stayed realistic.

6 Classification Models

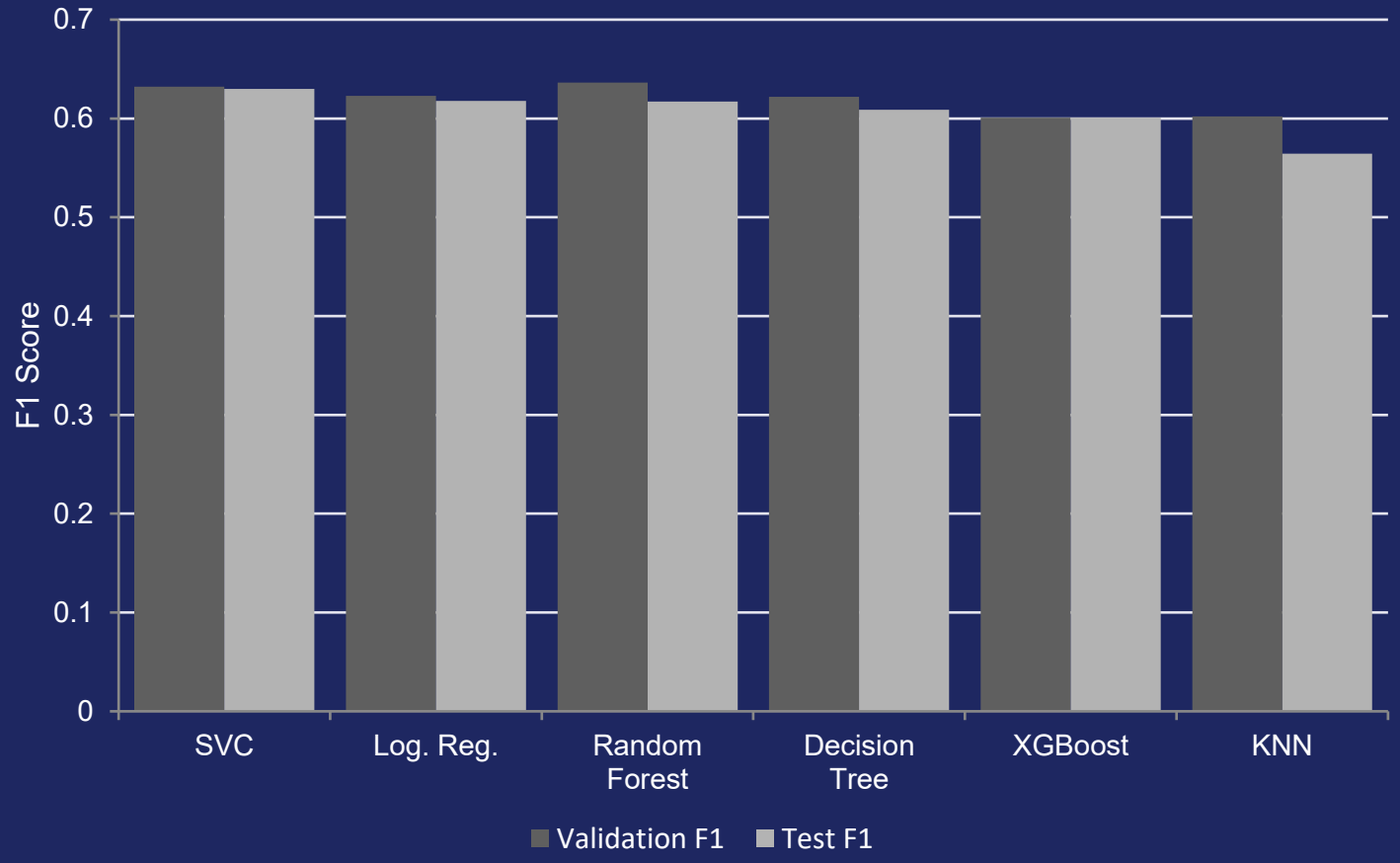
Logistic Regression, Decision Tree, Random Forest, XGBoost, KNN, and SVC — each scored on Accuracy, Precision, Recall, F1, and ROC-AUC.

Data Split



- Train 70%
- Validation 15%
- Test 15%

Model Comparison — F1 Score



XGBoost

Highest precision of any model, but the weakest recall, with 92% training accuracy vs. 76% on val/test, pointing to mild overfitting. This led to a very low F1 score

KNN

Least stable model overall: the largest drop from validation to test, most likely due to one hot encoding. As this model uses distance calculations, which with a lot of sparse dimensions, can cause low accuarcy

Most models clustered in the low to mid 0.60s on F1; SVC and Random Forest were the best

Verdict: Random Forest



82%

Recall on validation — the highest churn-catch rate of any model tested

Even though SVC scored slightly higher on test F1, I picked random forest because it has stronger recall, catching more at risk customers before they leave, which matters way more than raw accuracy.

Even a customer with only a 10% predicted churn risk is still worth flagging.